SAFE-SEARCH: Safety, Accountability, Faithfulness, and Evaluation in Conversational Search

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Abstract. Modern conversational search extends traditional IR into multi-turn interaction, where systems maintain context, infer evolving intent, reformulate queries, and increasingly invoke tools or take agentic actions. These capabilities create new guardrail needs across the search process. Intent may become adversarial over multiple turns. Retrieved content can steer downstream generation or propagate earlier errors. Systems must also decide when to answer, clarify, correct, abstain, or refuse. We propose **SAFE-SEARCH: Safety, Accountability, Faithfulness, and Evaluation in Conversational Search**, organized around four dimensions. **Safety** covers harmful and adversarial interactions, including evolving malicious intent, jailbreaks, and unsafe agent behavior. **Accountability** addresses privacy, bias, transparency, and responsible treatment of users and their data over time. **Faithfulness** concerns reliable retrieved information and grounded responses that do not carry errors across turns. **Evaluation and effective intervention** focuses on how guardrails should be assessed and how systems should respond to uncertainty or risk. SAFE-SEARCH will bring together researchers from IR, conversational search, NLP, recommender systems, and security to develop a shared taxonomy of multi-turn failure modes, identify evaluation and benchmark gaps, and outline priorities for future guardrail research and shared tasks.

Keywords: Conversational Search · Guardrails · Trustworthy Information Retrieval · Retrieval-Augmented Generation · Robustness · Responsible AI · Evaluation

1 Introduction and Motivation

Conversational search changes the interaction between users and information retrieval systems [21, 22]. Instead of issuing an isolated query and inspecting a ranked list, users can progressively express, refine, and reformulate their information needs through multi-turn dialogue [1]. Retrieval-Augmented Generation (RAG) systems [2] increasingly expose models to potentially untrusted external collections, while tool-using and agentic systems can translate generated decisions into new searches, API calls, or recommendations. As a result, adversarial influences are no longer necessarily confined to a single response. They can alter subsequent retrieval, persist through conversational memory, influence later decisions, or propagate into external actions [3, 4]. Safety must therefore be assessed not only at the level of individual prompts or responses, but across the *evolving information-seeking interaction* as a whole.

These systems are increasingly being deployed in mainstream enterprise applications, making these risks more acute. Real-world systems may operate over sensitive user queries, histories, and preferences, while their responses can influence consequential decisions. Failures can therefore extend beyond poor retrieval quality to privacy violations, harmful or discriminatory outcomes, misleading advice, security incidents, and misuse of user data, with potential regulatory and ethical consequences for system providers. Effective guardrails are thus becoming an important requirement for deploying conversational search systems responsibly in practice. These risks also make intervention a central part of system design: when should a conversational search system answer directly, ask for clarification, retrieve additional information, verify a claim, correct an earlier response, abstain, or refuse? Together, these challenges motivate guardrails that address safety, accountability, faithfulness, and the evaluation of effective interventions throughout the conversational search process. Guardrails must therefore move beyond isolated input and output checks to monitor and safeguard each stage of the evolving multi-turn interaction.

Existing work addresses important parts of this problem through general-purpose LLM guardrails [5–7] and a growing body of IR research on safer, more reliable, and trustworthy information access. Within IR, recent work across major venues has examined reliable query reformulation [16], prediction of harmful retrieval outcomes [17], search interventions for information verification [18], faithful clarification in conversational search [19], and verification of factual claims in real-world conversations [20]. These efforts address important pieces of the problem. SAFE-SEARCH brings them together around a shared question:

> *How should a conversational search system behave safely and reliably as the user's information need, retrieved evidence, conversational context, and system actions evolve over multiple turns?*

The workshop treats the *conversation*, rather than a single prompt, retrieved document, or generated response, as the primary unit of safety analysis.

2 Relevance to ECIR

SAFE-SEARCH lies primarily at the intersection of two ECIR 2027 research themes: *Conversational search and recommender systems* and *Societally-motivated IR research* [13]. It also connects naturally to ECIR's themes on RAG, large language models and agentic IR, evaluation, explainability, and accountability.

Conversational search research traditionally emphasizes dialogue quality, effective information access across turns, personalization, and user engagement. Societally motivated IR addresses complementary concerns including trustworthy IR, misinformation, harmful content, privacy, fairness, and ethics. SAFE-SEARCH brings these perspectives together by asking how safety, faithfulness, and accountability can be embedded into conversational search without diminishing utility and user satisfaction. A safe conversational system must still support high-quality dialogue. Rather than treating safety as a separate constraint that competes with relevance, utility, or engagement, we view it as a design objective that should be optimized alongside them. A central goal of SAFE-SEARCH is therefore to solicit methods and foster discussion on how guardrails can improve safety while preserving, and ideally enhancing, system effectiveness.

The workshop is also complementary to ECIR's IR-for-Good track. While IR-for-Good provides a broad home for societally motivated IR research, SAFE-SEARCH offers a focused forum for challenges that arise specifically in multi-turn interactive systems. Its emphasis is on how risks develop over the course of a dialogue, how earlier context influences later system behavior, and how safeguards can be integrated without undermining relevance or engagement. This focused scope also creates space for red-teaming studies, deployment lessons, incident analyses, benchmark design, and position papers.

3 Workshop Goals and Outcomes

SAFE-SEARCH has five primary objectives:

1. **Establish conversational-search safety as an end-to-end IR problem.** Move beyond isolated prompt and response moderation toward studying safety across interaction, retrieval, memory, and agentic actions.
2. **Advance evaluation practices.** Identify how multi-turn safety, failure propagation, recovery, and refusal trade-offs should be evaluated.
3. **Bridge research and real-world deployment.** Create a forum where academic research is informed by production failures, incident analyses, red-teaming findings, and deployment lessons from conversational systems at scale, while industry practitioners can draw on emerging methods, evaluation frameworks, and insights from the research community.
4. **Connect fragmented research communities.** Bring together researchers and industry practitioners from conversational IR, RAG, recommender systems, trustworthy IR, NLP, and security.

5. **Develop a shared research agenda.** Build an initial taxonomy of conversational-search failure modes and guardrail interventions, identify benchmark and evaluation gaps, and define priority problems that can support future datasets, collaborative resources, and shared tasks.

The workshop will target several concrete outcomes. Accepted archival papers will form workshop proceedings. During the panel discussion, participants will jointly develop a seed taxonomy of conversational-search failures spanning intent understanding, retrieval, conversational context, generation, and intervention, and map these failures to possible guardrail interventions and evaluation criteria. We will also identify priority benchmark and dataset gaps, deployment challenges, and candidate tasks that could form the basis of a future shared task. These artifacts will be consolidated into a public SAFE-SEARCH research agenda and released online after the workshop. Following the ECIR workshop call [14], we also intend to prepare a post-workshop report for the ACM SIGIR Forum and use these outcomes to guide future editions of the workshop.

4 Topics of Interest

SAFE-SEARCH invites work across four complementary themes aligned with the SAFE framework, with an emphasis on risks and interventions that emerge across multi-turn interactions.

Safety and Robustness.

- **Evolving harmful intent:** harmful requests, jailbreaks, prompt injection, and adversarial behavior that develop or compound across turns.
- **Context and memory risks:** failures introduced by dialogue history, persistent memory, cross-session information leakage, or accumulated context.

Accountability and Responsible Interaction.

- **Personalization and privacy over time:** responsible use of evolving user profiles, preferences, and conversational history.
- **Sycophancy, over-reliance, and user autonomy:** cumulative influence across dialogue, including systems that increasingly reinforce user beliefs or preferences rather than remain grounded in evidence.

Faithfulness and Reliability.

- **Grounding across turns:** ensuring claims, corrections, and citations remain supported as evidence and conversational context evolve.
- **Retrieval and source reliability over time:** handling conflicting, outdated, manipulated, or changing evidence across successive searches and responses.

Evaluation and Effective Intervention.

- **Multi-turn intervention strategies:** when to clarify, verify, re-search, correct, abstain, refuse, or restrict tool and agent actions as a dialogue evolves.
- **Evaluation and deployment:** multi-turn benchmarks, red teaming, reliability of LLM-based judges, incident analyses, deployment lessons, and negative results.

5 Workshop Format

SAFE-SEARCH is proposed as a **half-day, in-person workshop**. The workshop will combine invited and contributed presentations with structured interactive discussion.

5.1 Submissions and Proceedings

We plan to solicit the following contribution types:

- **Original Research papers**, in short (up to 6 pages) or full-length (up to 12 pages) formats, presenting completed research on the design, behavior, evaluation, or use of conversational search systems. Accepted papers may be presented orally or as posters, as determined by the Program Committee.
- **Demo and system papers** (up to 6 pages), presenting implemented guardrail systems, evaluation frameworks, datasets, benchmarks, or tooling, with an emphasis on system design and technical capabilities.
- **Experience reports** (up to 6 pages), documenting lessons from building or evaluating conversational search systems, including incident analyses, production failures, red-teaming findings, and operational trade-offs.
- **Position and vision papers** (up to 6 pages), articulating research challenges, conceptual frameworks, taxonomies, emerging problem formulations, or well-supported positions intended to stimulate discussion.

Page limits exclude references. Submissions will be peer reviewed by at least two members of the Program Committee. At least one author of each accepted contribution will be expected to attend and present. We plan to pursue archival publication through CEUR Workshop Proceedings, subject to eligibility.

5.2 Interactive Discussion

A subset of accepted contributions will be selected for short oral presentations. The program will devote time to panel discussions around questions such as:

- What should a taxonomy of conversational-search safety failures contain?
- What should multi-turn safety benchmarks measure beyond existing single-turn benchmarks?
- When should a system answer, clarify, retrieve again, verify, correct itself, abstain, or refuse?

- How should safety be balanced against relevance, usefulness, latency, and user autonomy?
- Which failures observed in deployed systems are poorly represented in current academic benchmarks?

The workshop requires only a standard conference room with presentation facilities, poster boards, and seating that can support group discussion.

6 Tentative Schedule

Table 1 shows a tentative morning program. The same structure can be shifted to an afternoon slot depending on the ECIR schedule.

7 Proposed Keynote Speakers

Dr. Emine Yilmaz, Professor of Computer Science at University College London, has agreed to deliver the academic keynote at SAFE-SEARCH. She leads UCL's Web Intelligence Group and has held senior roles in the IR community, including PC Chair for ECIR 2020 and SIGIR 2018. Her work in information retrieval, conversational search, and evaluation directly aligns with SAFE-SEARCH's focus. We also plan to invite one keynote speaker from industry.

8 Relationship to Existing Workshops

SAFE-SEARCH will be a first-edition workshop with *guardrails for multi-turn conversational search* as its primary research focus. Its core question is how safety risks emerge, accumulate, and propagate throughout the interaction process. The ECIR 2026 Workshop on Conversational Search for Complex Information Needs examined conversational systems for exploratory and multi-step information needs [8]; SAFE-SEARCH focuses specifically on how such interactions can fail, how risks propagate across turns, and when systems should intervene. Complementary ECIR workshops address credible retrieval and misinformation [9], longitudinal verification of claims and evidence [10], trustworthy

Table 1. Tentative half-day SAFE-SEARCH schedule.

Time	Activity
09:00–09:10	Welcome and workshop goals
09:10–09:40	Invited keynote I (Academic)
09:40–10:10	Spotlight talks from accepted contributions
10:10–10:30	Break
10:30–11:15	Poster session
11:15–11:45	Invited keynote II (Industry)
11:45–12:30	Invited panel and community discussion

scholarly information access [11], and broader changes to IR driven by generative AI [12]. In the adjacent NLP community, LLMSEC [15] focuses on threats and defenses for large language models, including prompt injection and guardrails.

9 Organizing Committee

9.1 Shasvat Desai

Shasvat Desai is a Staff Scientist on the Sponsored Search team at Walmart, where he works on information retrieval and LLM systems for e-commerce search. His research spans query and item understanding, intent-aware retrieval, relevance modeling and evaluation, and the use of LLMs for training and evaluating search systems. Previously, he worked on computer vision and multimodal systems for retail. He received his M.S. in Computer Science from UMass Amherst, and his research has been published at venues including SIGIR and CVPR.

9.2 Utshab Kumar Ghosh

Utshab Kumar Ghosh is a Ph.D. student in Computer Science at Missouri University of Science and Technology, working on information retrieval, evaluation, agentic search, and RAG. He previously spent three years as a machine learning engineer building production RAG systems. His first-author work has been accepted at ACM SIGIR, ICTIR, and CIKM. His research targets the gap between what looks relevant and what actually helps, across rankers and agents.

9.3 Ozel Yilmazel

Ozel Yilmazel is a PhD student at University of Massachusetts, Amherst, advised by Hamed Zamani and James Allan, where he also earned his B.S. and M.S. in computer science as a Bay State Fellow. His research develops retrieval for search agents and evaluates their faithfulness and behavior under uncertainty. His work on personalization of large language models appeared at ACM SIGIR 2026.

9.4 Rheeya Uppaal

Rheeya Uppaal is a final-year PhD candidate at the University of Wisconsin–Madison, advised by Junjie Hu. Her research focuses on reliable language models, with an emphasis on safety and faithfulness. She completed her M.S. in Computer Science at UMass Amherst, where she worked with Andrew McCallum. She also co-organized a 2024 Carpentries workshop on Trustworthy AI. She has published in ACL, COLM, ICLR, and EACL on topics spanning fairness and explainability.

9.5 Vijeta Deshpande

Vijeta Deshpande is a final-year Ph.D. student at University of Massachusetts Lowell supervised by Prof. Anna Rumshisky. He works on investigating and improving LLMs' capability to understand and generate diverse lexicons/words. His work has been published in ACL, EMNLP, and LREC-COLING

9.6 Keigo Sakurai

Keigo Sakurai is an Assistant Professor at the Faculty of Information Science and Technology, Hokkaido University, Japan. He received his Ph.D. in Information Science from Hokkaido University in 2025. His research focuses on recommender systems and data mining. His work has been published at major venues including SIGIR, RecSys, WSDM, CIKM, and ECIR.

9.7 Utkarsh Porwal

Utkarsh Porwal is a Senior Manager at Apple. His work has appeared in SIGIR, ECIR, NeurIPS, and SPIRE, with research spanning query understanding, ranking, and retrieval. He has also co-organized the SIGIR Workshop on eCommerce (ECOM-2019) and the KDD Converse 2020 Workshop

9.8 Shubham Chatterjee

Shubham Chatterjee is an Assistant Professor of Computer Science at Missouri University of Science and Technology, where he leads the IRIS Group. His research focuses on RAG, LLMs, and conversational information access. His work has appeared in SIGIR, ECIR, CIKM, EMNLP, and ICTIR. He received his Ph.D. from the University of New Hampshire under Laura Dietz and previously worked with Jeff Dalton at the Universities of Glasgow and Edinburgh. He has co-organized tutorials on neuro-symbolic information retrieval at ECIR 2023 and SIGIR 2023, ProActLLM workshop at CIKM 2025, GRAIL workshop in CIKM 2026, and has received outstanding reviewer awards from ECIR and CIKM.

9.9 Tracy Holloway King

Dr. Tracy Holloway King is a Senior Principal Scientist focusing on multimodal search science in Adobe's Search & Discovery team. After getting her Ph.D. from Stanford, she was an NLP researcher in PARC for over a decade. She then moved to applied science roles at Microsoft Bing, eBay Search Science, and Amazon Query Understanding before joining Adobe. She has co-organized workshops at SIGIR (ECOM workshop 2020-2026), CIKM (MMSR 2024), ICDM (MMSR 2025), and co-organized the Industry Track at SIGIR 2023 and CIKM-2026.

9.10 Debasis Ganguly

Debasis Ganguly has been working as an assistant professor in the University of Glasgow, UK since 2021. He was a former research scientist in IBM Research, Ireland, where he worked on information extraction and fact verification of scientific claims. His general research interest is directed towards retrieval models and RAG, predicting performance of AI models, and developing methodologies for fair, interpretable, trustworthy and privacy-preserving AI. Over the years he has published over a hundred research papers in top tier conferences and journals, namely SIGIR, CIKM, ECIR, TOIS, IPM etc.

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